Stage 2: Measles Model Parameters, Population Immunity, and Critical Vaccination Coverage

Q1. Why is the latent period 10 days and not 14?

Step 1 - Definitions. Incubation is the time from infection to symptoms (here, rash). Latency is the time from infection until a person becomes infectious. These are not interchangeable: measles transmission can begin before the rash.

Step 2 - Clinical timing and calculation. The draft reports rash onset about 14 days after exposure and infectiousness from four days before to four days after rash onset (Section 3.2).

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Latent period = 14 - 4 = 10 days

1/sigma = 10 days; sigma = 1/10 = 0.100 day^-1

Infectious period = 4 + 4 = 8 days

1/gamma = 8 days; gamma = 1/8 = 0.125 day^-1
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Step 3 - Effect of incorrectly using 14 days. Setting 1/sigma = 14 gives sigma = 1/14 = 0.07143 per day instead of 0.100 per day. The model transitions E to I according to dI_k/dt = sigma E_k - gamma I_k. A smaller sigma prolongs the noninfectious exposed stage, generally slowing epidemic growth and shifting the epidemic peak later.

This does **not** change the reproduction number under the stated model assumptions. Section 2.4, Step 3, reduces the next-generation matrix to K = B/gamma and defines R0 = beta/gamma; sigma cancels because all exposed individuals eventually become infectious. The timing of transmission changes, not the expected total number of secondary infections.

Answer. The latent period is 10 days (14 minus 4), because infectiousness begins before rash onset. A 14-day latent period would delay the simulated epidemic without changing R0.

Q2. Why cannot kindergarten coverage be used as s1?

Step 1 - Apply the susceptibility equation. Group 1 is the general population of all ages, whereas kindergarten coverage describes children of approximately five years of age. Equation (3) gives s1 = 1 - eps1*c11 - eps2*c21. Under the simplifying assumption of two-dose coverage only, s1 = 1 - eps2*c21, with eps2 = 0.97 and R0 = 15.

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West Texas: s1 = 1 - 0.97(0.932) = 0.09596; 15*s1 = 1.43940

Upstate South Carolina: s1 = 1 - 0.97(0.889) = 0.13767; 15*s1 = 2.06505

Short Creek: s1 = 1 - 0.97(0.790) = 0.23370; 15*s1 = 3.50550
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Outbreak	Kindergarten coverage	s1	R0*s1
West Texas	93.2%	0.09596	1.44
Upstate South Carolina	88.9%	0.13767	2.07
Short Creek	79.0%	0.23370	3.51

Step 2 - Apply Theorem 1. With theta = 1, K = diag(R0*s1, R0*s2), so Rv = max(R0*s1, R0*s2). Theorem 1 requires Rv < 1 for epidemic decline. Each kindergarten-derived R0*s1 exceeds 1, implying that the general population could sustain

transmission independently if isolated.

Step 3 - Compare with observed cases. Instead, approximately 90% of South Carolina cases occurred within the under-vaccinated community; about 97% of early Texas cases had no documented vaccination. The case concentration is inconsistent with treating the entire general population as having kindergarten-level susceptibility. Adult immunity is generally higher than the immunity inferred from this narrow school-age sample.

Step 4 - Serology. National measles seropositivity was 92%, yielding the conservative bound s1 <= 1 - 0.92 = 0.08, and R0*s1 <= 15(0.08) = 1.20. The upper bound can still exceed 1 because antibody assays do not capture all protected individuals; absence of detectable antibodies is not identical to absence of immune protection.

Step 5 - Stage 3 estimation. Accordingly, the study estimates s1 over [0, 0.08], using the case distribution between the general population and the under-vaccinated community (especially the approximately 90% community share in South Carolina), alongside outbreak time series and mixing assumptions. The split is identifying information, not by itself a unique estimate.

Answer. Kindergarten vaccination data measure a restricted age cohort rather than population-wide effective susceptibility. Using them as s1 produces unrealistic general-population transmission predictions, so s1 must be estimated from outbreak data.

Q3. Why do all critical-coverage curves end at 96.2% when theta = 1?

Figure note. The critical vaccination coverage curves are labeled Figure 3 in the uploaded draft; Figure 2 shows weekly case counts.

Step 1 - Corollary 1. For a = R0*s1*[theta + (1-theta)*p1], the critical community susceptibility is

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s2* = (1-a) / {R0*[theta+(1-theta)*p2]*(1-a)
+ R0^2*s1*(1-theta)^2*p1*p2}
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At theta = 1, (1-theta) = 0, so provided a < 1,

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s2* = (1-a)/[R0*(1-a)] = 1/R0 = 1/15.
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Step 2 - Convert to two-dose coverage. Since s2 = 1 - eps2*c2, the critical two-dose coverage is

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c2* = (1 - 1/R0)/eps2
= (1 - 1/15)/0.97 = (14/15)/0.97
= 0.9621993... = 96.22%.
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To ensure Rv < 1, actual coverage must be strictly greater than the threshold. At theta = 1 the community is isolated, so its threshold does not depend on s1 or the community population share.

Step 3 - Why Short Creek can reach zero at low theta. The population shares are

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Short Creek: p2 = 3605/(511497+3605) = 0.0069986 ~= 0.70%
West Texas: p2 = 5000/(23956+5000) = 0.1726758 ~= 17.27%
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For random mixing (theta = 0), Rv = R0*(p1*s1 + p2*s2). At zero community vaccination, s2 = 1. Using illustrative s1 = 0.04:

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Short Creek: Rv = 15*(0.9930014*0.04 + 0.0069986)
= 0.7008 < 1
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West Texas: Rv = 15*(0.8273242*0.04 + 0.1726758)

= 3.0865 > 1
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Short Creek is so small relative to the surrounding immune population that, under sufficiently random mixing, the population-wide reproduction number can fall below one even with no community vaccination. In West Texas, the community is large enough that Rv exceeds one at theta = 0 even if s1 = 0: 15(0.1726758) = 2.5901 > 1. Thus its critical community coverage remains positive throughout the mixing range.

Answer and public-health meaning. Every valid curve approaches 96.2% because isolation makes the community responsible for its own herd-immunity threshold, irrespective of outside immunity or population size. Highly clustered communities need very high local two-dose coverage to prevent sustained transmission.